

Kernel Composition Report

6 February 2026

Overview

This report summarizes kernel composition experiments and compares them to prior runs. The goal is to evaluate whether composing more kernels improves performance and also and to visualize the resulting composite filters.

Composition configurations

Run	total number of kernels	comp. pool	Test AUC	Test ACC
A	90	25	0.8662	0.7968
B	450	125	0.8666	0.7920
C	2250	625	0.8869	0.8107
D	4500	1250	0.8933	0.8150

Kernel visualization (composite filters)

Each image below shows the learned composite kernel (red/blue indicate positive/negative response regions). Stronger contrast suggests a more selective filter, while broader smooth regions indicate texture/low-frequency emphasis.

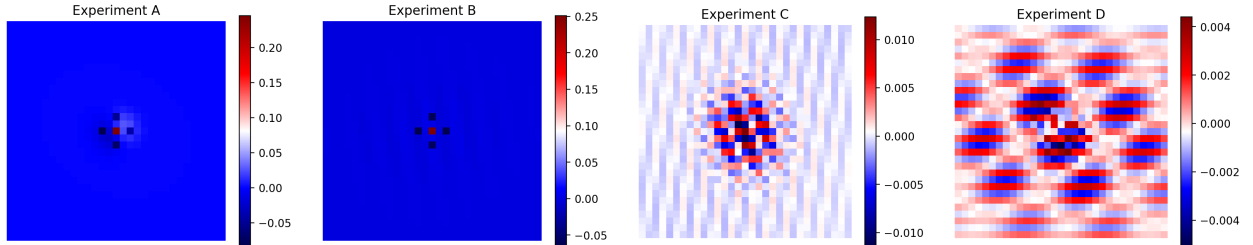


Figure 1: Composite kernels for A, B, C, and D (left to right).

Comparison with prior experiments

Compared with earlier baselines (best test AUC 0.9239 from `exp_022_mmr` and best test ACC 0.8182 from `exp_025_mmr`), run D yields the strongest composite test accuracy (0.8150) and the highest composite test AUC (0.8933). ABESS runs peak at test AUC 0.9236 and ACC 0.8032, which are below the composite accuracy of run D.

How parameter changes affect quality

- Increasing the composition pool size ($25 \rightarrow 125 \rightarrow 625 \rightarrow 1250$) steadily improves composite AUC/ACC.
- Larger total kernel banks provide more diverse filters for composition, which improves composite stability and accuracy.
- The largest pool (D) delivers the best composite test accuracy and AUC among composition runs.

Composition vs non-composition accuracy

Run	Test ACC (regular)	Test ACC (composite)
A	0.8487	0.7968
B	0.8503	0.7920
C	0.8492	0.8107
D	0.8733	0.8150

Observation: many kernels vs single composite

Based on the previous runs, the single composite-kernel approach is preferable. It outperforms the large multi-kernel baselines while remaining simpler, faster, and easier to deploy.

Conclusion

Run D (500 kernels per family with a 1250-kernel composition) is the best overall. It achieves the highest composite test AUC (0.8933) and composite test accuracy (0.8150), and it also delivers the best regular-model test accuracy (0.8733). Composition improves interpretability and consistency, while the regular model maintains a higher ceiling on accuracy.

Next steps

- Run more experiments with different parameter settings to further improve accuracy.
- Evaluate on a new dataset to confirm generalization.
- Study learning a kernel-generating model to produce strong kernels without random sampling.